Chase Chemero (SID: 20817851)
CS-560: Spring 2026
Final Project Proposal V1.3

Project Proposal: Analyzing Data Breach Trends

Basic Information

Project Title: Washington State Data Breach Analysis & Visualization
Name: Chase Chemero
Email: cchemero@usfca.edu
GitHub Repository: CS-560 Final Project (Chemero)

Background and Motivation

In an increasingly digital world, data privacy has become a paramount concern for citizens and legislators alike. Washington State requires organizations to report data breaches that affect its residents, creating a rich dataset for public analysis. My motivation for this project stems from a desire to understand the evolving landscape of cybersecurity. By visualizing this data, I hope to uncover whether data breaches are becoming more frequent, which industries are most vulnerable, and how long it typically takes for an organization to notify the public after a breach occurs.

Project Objectives

The primary goal of this visualization is to allow users to explore the severity and nature of data breaches in Washington.

Objective 1: Temporal Trends
Users will be able to identify if the frequency and scale (number of residents affected) of breaches are increasing over time.

Objective 2: Breach Categorization
Users will be able to determine which types of breaches (e.g., Hacking, Phishing, Unauthorized Access) are most prevalent.

Objective 3: Risk Assessment
Users will be able to identify which types of personal information (SSN, Health info, Financial data) are most frequently compromised.

Objective 4: Accountability Analysis
Users will be able to visualize the "Notification Gap" which is the time elapsed between the date of the breach and the date residents were notified.

Data

The project will utilize the dataset: Data Breach Notifications Affecting Washington Residents which is approximately ~6,800 rows and covers the 2017 to 2025 time range.

Format: CSV/JSON

Key Fields:

- Organization Name
- Breach Start/End Dates
- Date of Notification
- Number of WA Residents affected
- Type of Breach
- Information Compromised

Limitations & Considerations:

- The dataset is self reported, which may introduce reporting bias or inconsistencies.
- Some records contain missing or “Unknown” values, particularly in the “Number of Residents Affected” field.
- Date fields may be inconsistent or incomplete, requiring normalization before time series analysis.
- The dataset is limited to Washington State residents, which may affect generalizability to broader cybersecurity trends.

Data Processing

Cleaning: The dataset will require significant cleaning regarding date formats to ensure they are ISO compliant for time series analysis. I will also need to handle "Unknown" or missing values in the "Number of Residents" column, perhaps by imputing the median or labeling them as "Not Reported.”

Processing Pipeline:

Preprocessing Stage (Python):

- Data cleaning, normalization of date formats, and handling missing values
- Calculation of derived fields such as Notification Lag and Info Density
- Export of a cleaned and structured dataset for visualization

Visualization Stage (D3.js and JavaScript):

- Dynamic filtering and aggregation
- Real time interaction handling (e.g. brushing, category selection)
- Rendering of visual components and transitions

Derived Attributes:

Notification Lag: A calculated field representing Notification Date to Breach Start Date.

Category Aggregates: Grouping specific breach descriptions into broader categories (e.g., "Cyber Attack," "Human Error," "Physical Theft").

Information Density: A count of how many different types of personal data (SSN, DL, etc.) were stolen per incident.

Visualization Design

Design 1: The Temporal Dashboard
A focus on a large Interactive Area Chart showing the number of residents affected over the last 5 years, with a "brushing" feature to zoom into specific months.

Design 2: The Categorical Network
A node link diagram showing the relationship between "Organization Type" and "Information Stolen."

Design 3: The Heatmap View
A visualization focusing on when breaches occur and the "Notification Lag" intensity, helping to see if certain times of year are more prone to attacks.

Final Design: Integrated Dashboard
The final design will be a Dashboard Interface. It will feature a top level Line Chart for temporal trends (Objective 1). Below that, a Grouped Bar Chart will compare Breach Types (Objective 2). On the right, an interactive Donut Chart will display the breakdown of compromised data types (Objective 3). Selecting a bar or slice will filter the entire dashboard to show data specific to that category.

Justification:
This layout allows for high level overview and deep dive filtering, ensuring all four objectives are met within a single view without overwhelming the user.

Must-Have Features

Interactive Timeline: Users can filter data by year/month. (Meets Obj 1)

Categorical Filter:
A way to toggle between breach types (Hacking vs. Phishing). (Meets Obj 2)

Search Functionality:
A search bar to look up specific organizations (e.g., “Zillow") to see their breach history. (Meets Obj 1 & 4)

Notification Delay Histogram:
A visualization showing the distribution of how long companies take to report breaches. (Meets Obj 4)

Project Schedule

Week 1: Research, Design, & Data Preprocessing

Deliverables:
- Five Design Sheet sketches
- Cleaned dataset

Tasks:
- **Apr 4:**
    - Finalize project scope and objectives
    - Review dataset structure and identify key variables
- **Apr 5–6:**
    - Perform Exploratory Data Analysis using Python
    - Identify inconsistencies in "Notification Date" and "Breach Start" fields
- **Apr 7–8:**
    - Clean dataset (handle nulls, normalize categories, standardize date formats)
    - Compute derived fields (Notification Lag, Info Density)
- **Apr 9:**
    - Create and refine Five Design Sheet sketches
    - Initialize GitHub repository and D3.js development environment

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Week 2: Frontend Scaffolding & Core Features

Deliverables:
- Revised Proposal, Related Work, and Website
- Functional dashboard layout
- Initial interactive charts

Tasks:
- **Apr 10:**
    - Revised Proposal, Related Work & Website
    - Submit revised proposal and related work documentation
    - Begin HTML/CSS layout (dashboard structure, chart containers)
- **Apr 11:**
    - Continue HTML/CSS layout; publish project website
- **Apr 12–13:**
    - Implement Interactive Timeline (area/line chart)
    - Bind cleaned dataset to visualization
- **Apr 14–15:**
    - Develop categorical visualizations (bar chart / donut chart)
    - Implement filtering logic across components
- **Apr 16:**
    - Validate interactivity and ensure consistent data updates across views

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Week 3: Advanced Interactivity & Visual Polishing

Deliverables:
- Search functionality
- Notification delay histogram

- Polished UI/UX
- Alpha Release

Tasks:
- **Apr 17:**
    - Alpha Release
    - Package and submit Alpha Release (functional dashboard with core charts)
    - Begin search/autocomplete functionality for organizations
- **Apr 18:**
    - Complete search/autocomplete implementation
- **Apr 19–20:**
    - Build Notification Delay Histogram
    - Validate correctness of lag calculations
- **Apr 21–22:**
    - Refine visual encodings (color schemes, labels, tooltips)
    - Improve responsiveness and layout
- **Apr 23:**
    - Conduct usability testing and incorporate feedback

Week 4: Deployment, QA, & Final Deliverables

Deliverables:
- Beta Release

Tasks:
- **Apr 24–25:**
    - Perform QA testing (cross-browser compatibility, edge cases)
- **Apr 26–27:**
    - Finalize documentation (GitHub README, architecture overview)
- **Apr 28–29:**
    - Begin final report (design decisions, outcomes, limitations)
- **Apr 30:**
    - Beta Release
    - Submit Beta Release

Week 5: Presentation Preparation

Deliverables:
- Project Presentation

Tasks:
- **May 1–4:**
    - Build presentation slides covering design rationale, key findings, and live demo
- **May 5–9:**
    - Rehearse presentation; incorporate advisor feedback
- **May 10:**
    - Final dry run; polish slides and demo flow
- **May 11:**
    - Submit/deliver Project Presentation

Week 6: Report Draft

Deliverables:

- Project Report Draft

Tasks:

- **May 12–14:**
  - Write report sections: Introduction, Related Work, Design Decisions
- **May 15–16:**
  - Write Outcomes, Limitations, and Future Work sections
- **May 17:**
  - Submit Project Report Draft

Week 7: Final Submission

Deliverables:

- Final Submission

Tasks:

- **May 18:**
  - Incorporate feedback from report draft; record video demo
- **May 19:**
  - Final polish; dashboard, report, and video
- **May 20:**
  - Submit Final Submission package

Works Cited

Data Sources

- U.S. General Services Administration. (n.d.). Data breach notifications affecting Washington residents. Data.gov. https://catalog.data.gov/dataset/data-breach-notifications-affecting-washington-residents-personal-information-breakdown

Visualization & Design Methodology

- Five Design Sheet Methodology. (n.d.). The Five Design Sheet methodology. https://fds-design.github.io/
- Roberts, J. C., Headleand, C., & Ritsos, P. D. (2016). Sketching designs using the Five Design Sheet methodology. IEEE Transactions on Visualization and Computer Graphics.

Tools & Technologies

- D3.js. (n.d.). Data-Driven Documents. https://d3js.org/
- Pandas. (n.d.). Pandas documentation. https://pandas.pydata.org/

Domain Context (Cybersecurity & Data Breaches)

- Identity Theft Resource Center. (2023). Annual data breach report. https://www.idtheftcenter.org/
- Verizon. (2023). Data breach investigations report (DBIR). https://www.verizon.com/business/resources/reports/dbir/

AI Usage Statement

This project utilized AI assisted tools, including Gemini, to support formatting, organization, and refinement of written content. All substantive ideas, design decisions, data analysis plans, and implementation details were developed independently by the author. AI tools were used solely to improve clarity and presentation.